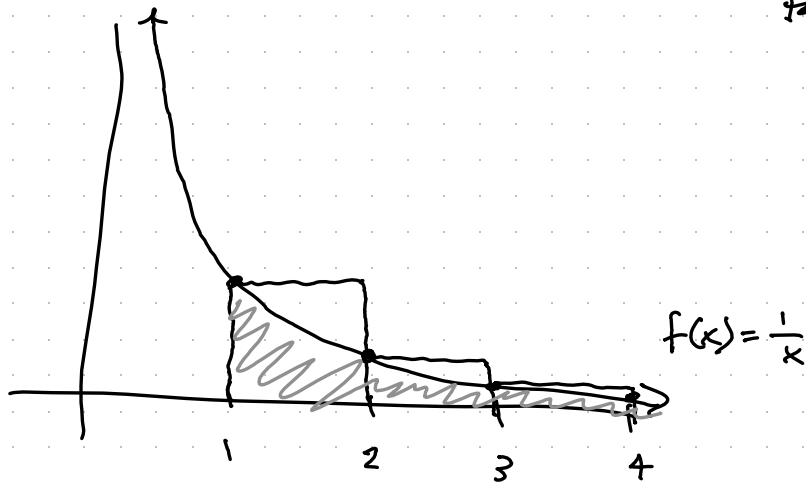


Integral test - 5.3

Example: The harmonic series diverges.

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty.$$

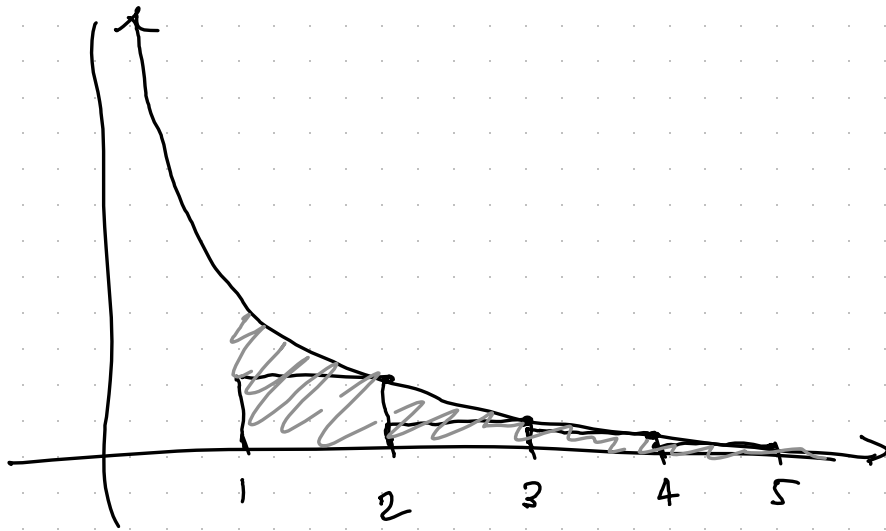
$\sum_{n=1}^{\infty} \frac{1}{n}$ is an overestimate
for $\int_1^{\infty} \frac{1}{x} dx$.



Recall: We showed
that $\int_1^{\infty} \frac{1}{x^p} dx$
converges if
and only if $p > 1$.

$$\int_1^{\infty} \frac{1}{x} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx = \lim_{t \rightarrow \infty} (\ln t) = \infty.$$

Example : $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.



→ area on this side is only $\sum_{n=2}^{\infty} \frac{1}{n^2}$

$\int_1^{\infty} \frac{1}{x^2} dx$ converges, so gray area is finite,

hence $\sum_{n=2}^{\infty} \frac{1}{n^2}$ and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converge as well.

Integral test: If $f(x)$ is a positive, continuous, decreasing function, then for $a_n = f(n)$, $\sum_{n=1}^{\infty} a_n$ and $\int_1^{\infty} f(x) dx$ either both converge or they both diverge.

Example: We said $\int_1^{\infty} \frac{1}{x^p} dx$ converges iff $p > 1$, so also $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges iff $p > 1$.
← sometimes called "p-series"

Note: $\int_1^{\infty} f(x) dx$ and $\sum_{n=1}^{\infty} a_n$ don't necessarily converge to the same value.

$$\int_1^{\infty} \frac{1}{x^2} dx = 1 \quad \text{but} \quad \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Example: $\sum_{n=1}^{\infty} \frac{1}{n^{2/3}}$ diverges; $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ converges.

Example: Does $\sum_{n=1}^{\infty} \frac{1}{\sqrt{2n-1}}$ converge or diverge?

$\frac{1}{\sqrt{2x-1}}$ is a positive, decreasing, continuous function,

so we can use the integral test.

$$\int_1^{\infty} \frac{1}{\sqrt{2x-1}} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{\sqrt{2x-1}} dx \quad \begin{array}{l} u = 2x-1 \\ du = 2 dx \end{array}$$

$$= \lim_{t \rightarrow \infty} \left(\frac{1}{2} \int_1^{2t-1} \frac{1}{\sqrt{u}} du \right) = \lim_{t \rightarrow \infty} \frac{1}{2} \cdot 2\sqrt{u} \Big|_1^{2t-1}$$

$= \lim_{t \rightarrow \infty} \sqrt{2t-1} - 1 = \infty$, and thus $\sum_{n=1}^{\infty} \frac{1}{\sqrt{2n-1}}$ diverges
by the integral test.